Sec 11 A 8.19 8.28 8.29 8.32 8.36

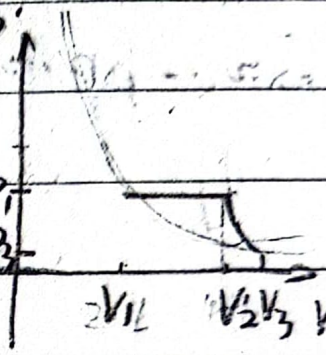
B 8.41 8.43 8.44

8.19 $\nu = 2mol$ $T_1 = 300K$ $V_1 = 20L$

$P_1V_1 = \nu RT_1 \Rightarrow P_1 = 2.439\times10^5 Pa$

$V_2 = 40L$, $\gamma = 1 + \frac{2}{i} = \frac{5}{3}$

$$\begin{cases} P_1V_2^{\gamma} = P_3V_3^{\gamma} \\ P_1V_1 = P_3V_3 \end{cases} \Rightarrow \begin{cases} P_3 = 4.3\times10^4 Pa \\ V_3 = 4\sqrt{2}V_1 = 113L \end{cases}$$



$$W = P_1(V_2 - V_1) + \int_{V_2}^{V_3} V^{-\frac{5}{3}} P_1V_2^{\frac{5}{3}}\,dV = P_1(V_2 - V_1) - \frac{3}{2}P_1V_2^{\frac{5}{3}}(V_3^{-\frac{2}{3}} - V_2^{-\frac{2}{3}})$$

$$\Rightarrow \frac{5}{2}P_1V_1 = 1.22\times10^4 J$$

(2) $Q = 1.22\times10^4 J$ (3) $\Delta E = 0$ (4) $W = 1.22\times10^4 J$ (5) $V_3 = 113L$

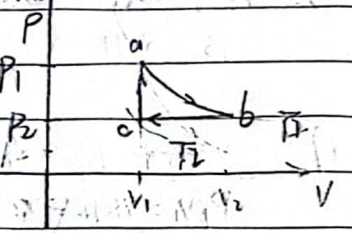
8.28 ab: $W = 0$ $Q = \Delta E = \frac{3}{2}\nu R\Delta T = 61J$

bc: $\Delta E = 0$ $Q = W = 200J$

$\Delta E = 61J$

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8.29 $\nu=0.1\,mol$ $P_1=1\times10^5Pa$ $V_1=3.2\times10^{-3}m^3$



ab: $Q_{吸}=W=\int_{V_1}^{V_2}\frac{1}{V}P_1V_1\,dV=\ln2P_1V_1$

$=221.8J$

bc: $-Q_{放}=\Delta E+W\Rightarrow Q_{放}=400J$

ca: $Q_{吸}=\Delta E=249J$

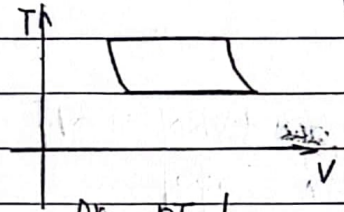
$\therefore\eta=\frac{Q_{吸}-Q_{放}}{Q_{吸}}=13.4\%$

8.32 ab: 吸热 bc: 放热

$T_1V_1^{\gamma}=T_2V_2^{\gamma}\Rightarrow T_2=T_1(\frac{V_1}{V_2})^{\gamma}$

$P_2V_2=\nu RT_2\Rightarrow P_2=\frac{\nu RT_1(\frac{V_1}{V_2})^{\gamma}}{V_2}$

不是



ab: $Q_{吸}=W=\int_{V_1}^{V_2}\frac{\nu RT_1}{V}dV=\nu RT_1\ln\frac{V_2}{V_1}$

bc: $Q_{放}=-\Delta E=\frac{1}{2}\nu R(T_1-T_2)$

ca: $Q=0$

$\therefore\eta=\frac{Q_{吸}-Q_{放}}{Q_{吸}}=\frac{T_1\ln\frac{V_2}{V_1}-\frac{1}{2}(T_1-T_2)}{T_1\ln\frac{V_2}{V_1}}$

8.36. $T_1=303K$, $T_2=573K$, $T_2'=263K$

$Q_{吸}=1.675\times10^7J$

$e_c=\frac{T_2}{T_1-T_2}=\frac{Q_{吸}}{W}\Rightarrow W=1.66\times10^6J$

$W=Q_{放}-Q_{吸}\Rightarrow Q_{放}=1.841\times10^7J$

$e_c=\frac{T_2'}{T_1-T_2'}=\frac{Q_{吸}}{W'}\Rightarrow W'=2.36\times10^6J$

$\Delta W=7\times10^5J$

8.41. $W=7300J$

$Q=\Delta E+W\Rightarrow Q=2800J$

$S=\int_0^d\frac{dQ}{T}+\int_d^c\frac{dQ}{T}$

$=\int_d^d\frac{\nu C_{v,m}dT}{T}+\int_d^c\frac{\nu C_{p,m}dT}{T}$

$=\nu C_{v,m}\ln\frac{T_d}{T_a}+\nu C_{p,m}\ln\frac{T_c}{T_d}=23.0J/K$

$Q_{v,m}=$

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$dQ = \nu C_{V,m} dT + pdv$ $\quad PV = \nu RT$

$dQ = cm\,dT$

8.43 $P_n = 101325\,Pa \quad T_1 = 273K \quad T_2 = 333K$

$\Delta S_1 = \frac{\nu M \lambda}{T_1} = 22.02\,J/K$

$\Delta S_2 = \int_{T_1}^{T_2} \frac{cM\,dT}{T} = cM \ln \frac{T_2}{T_1} = 14.97\,J/K$

$\Delta S = \Delta S_1 + \Delta S_2 = 36.99\,J/K$

8.44

$\Delta S_1 = \frac{dQ}{T} = \nu_1 R \ln \frac{V_2}{V_1} = \nu_1 R \ln 2 =$

A	B
m_1/g	$2m_1/g$

$\Delta S_2 = \nu_2 R \ln 2$,

$\nu_1 = \nu_2 = \frac{m_1}{M_1} = \frac{m_2}{M_2} \quad \Delta S = 2 \ln 2 R \frac{m_1}{M_1}$

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